

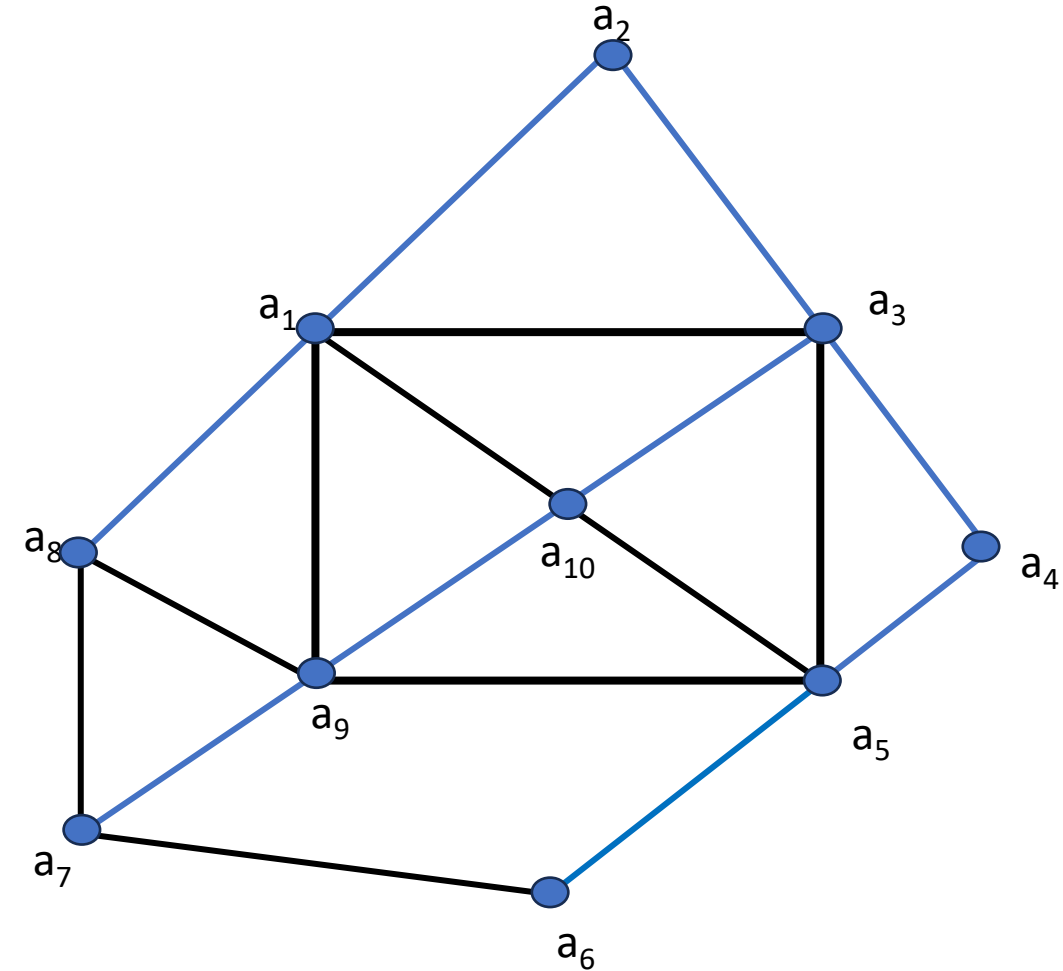
Minimal Spanning Tree

Group-5, Discrete Mathematics

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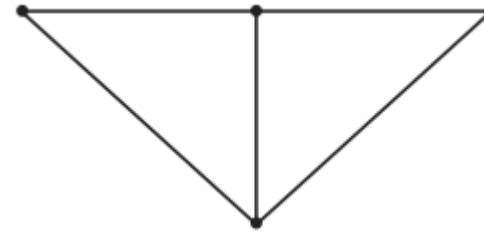
Spanning Tree

- A tree T is said to be a spanning tree of a connected graph G if T is a subgraph of G and T contains all vertices of G . A spanning tree of a graph is a subgraph that contains all the vertices and is a tree.
- For example, the subgraph in blue line is a Spanning tree of the graph



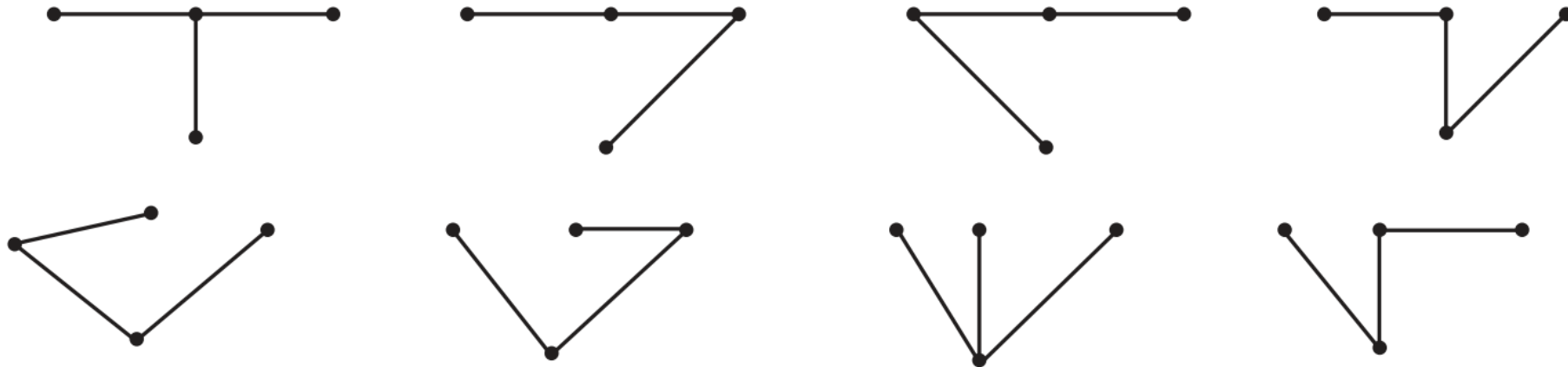
Example

- *Find all spanning trees of the graph G.*



SOLUTION. Graph G has 4 vertices and hence each spanning tree must have $4 - 1 = 3$ edges.

Thus, each tree can be obtained by deleting two of the 5 edges of G. This can be done in 10 ways except that two of the ways lead to disconnected graphs. Thus, there are 8 spanning trees as follows

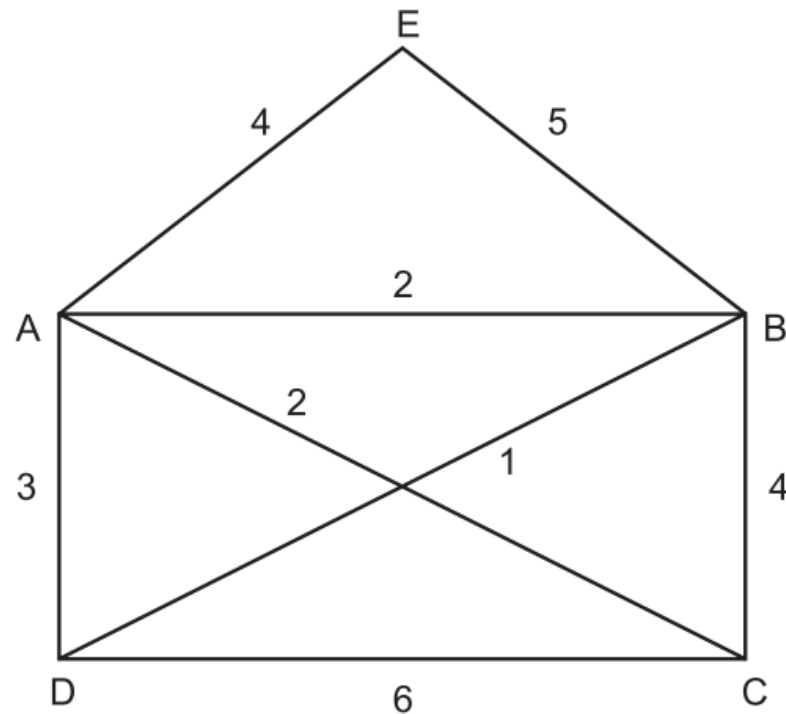


A Hamiltonian path is a spanning tree.

- We know that a Hamiltonian path P in a connected graph G is a path which contains every vertex of G and that if G has n vertices then P has $(n - 1)$ edge. Thus, P is a connected subgraph of G with n vertices and $n - 1$ edges. Therefore, P is a tree.
- Since P contains all vertices of G , it is a spanning tree of G .

Weighted Graph

A weighted graph is a graph G in which each edge e has been assigned a non-negative number $w(e)$, called the weight of e . The below figure shows a weighted graph



Minimal Spanning Tree

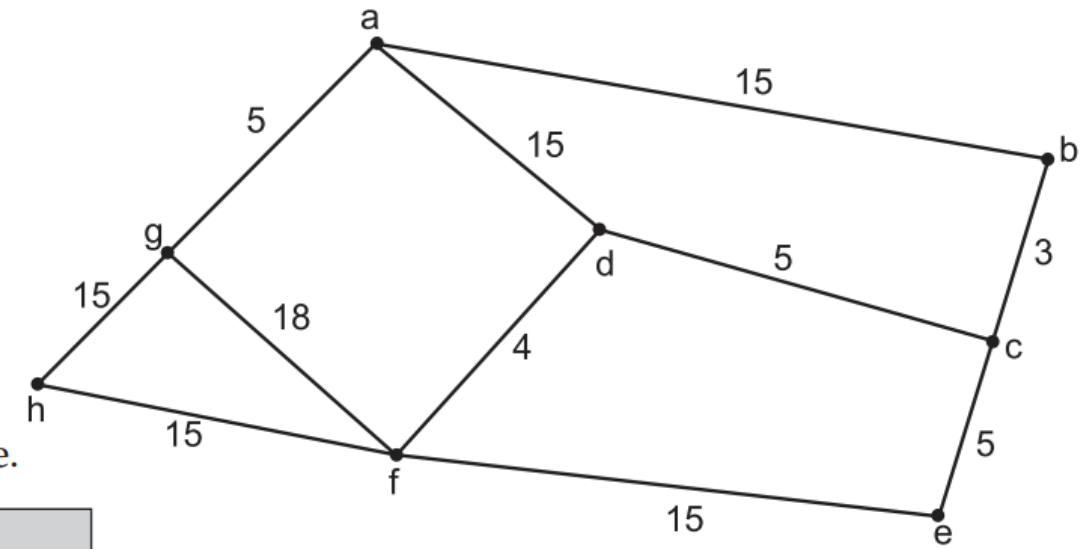
- Let G be a weighted graph. A minimal spanning tree of G is a spanning tree of G with minimum weight. If graph G is a weighted graph then the weight of spanning tree T of G is defined as the sum of the weights of all the branches in T .

Algorithm for minimal spanning tree

(A) Kruskal's algorithm :

- Kruskal's algorithm to find a shortest spanning tree in a given graph is as follows :
- 1. List all edges of the graph G in order of nondecreasing weight.
- 2. Select any edge of minimal value that is not a loop. If there is more than one edge of minimal value, arbitrarily choose one of these edges.
- 3. For each successive step select any remaining edge of G having minimal value that does not form a circuit with the edges already selected.
- 4. Continue step (3) until $(n - 1)$ edges have been selected and these edges will constitute the desired shortest spanning tree.

Use Kruskal's algorithm find a minimal spanning tree (MST) of the graph of figure.



SOLUTION. First we collect the edges with their weights into a table.

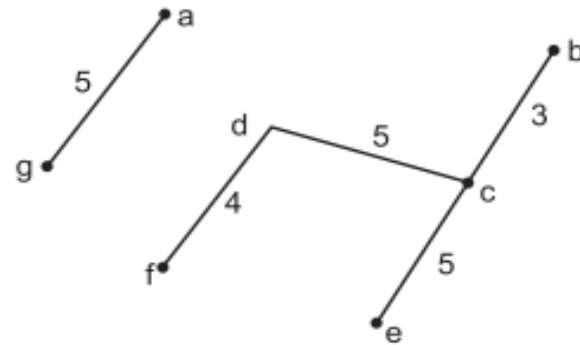
Edge	Weight
{b, c}	3
{d, f}	4
{a, g}	5
{c, d}	5
{c, e}	5
{a, b}	15
{a, d}	15
{f, h}	15
{g, h}	15
{e, f}	15
{f, g}	18

The steps of finding a minimal spanning tree as follows :

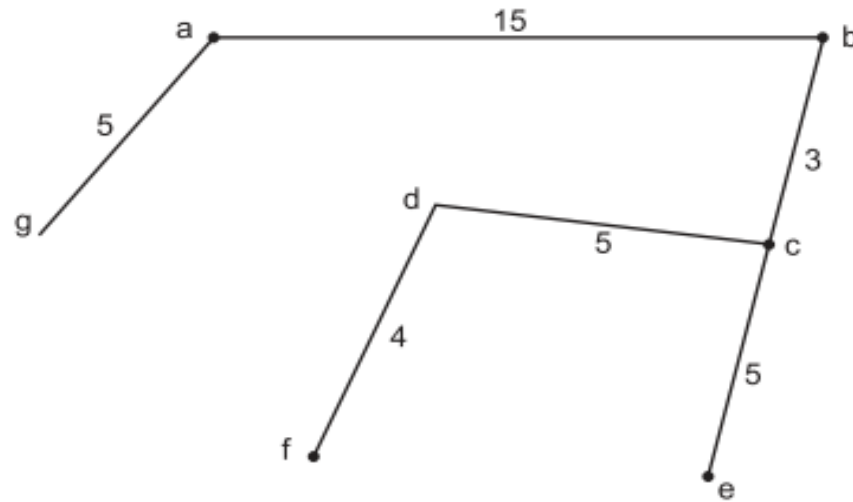
- (i) Choose the edge $\{b, c\}$ as it has minimal weight.



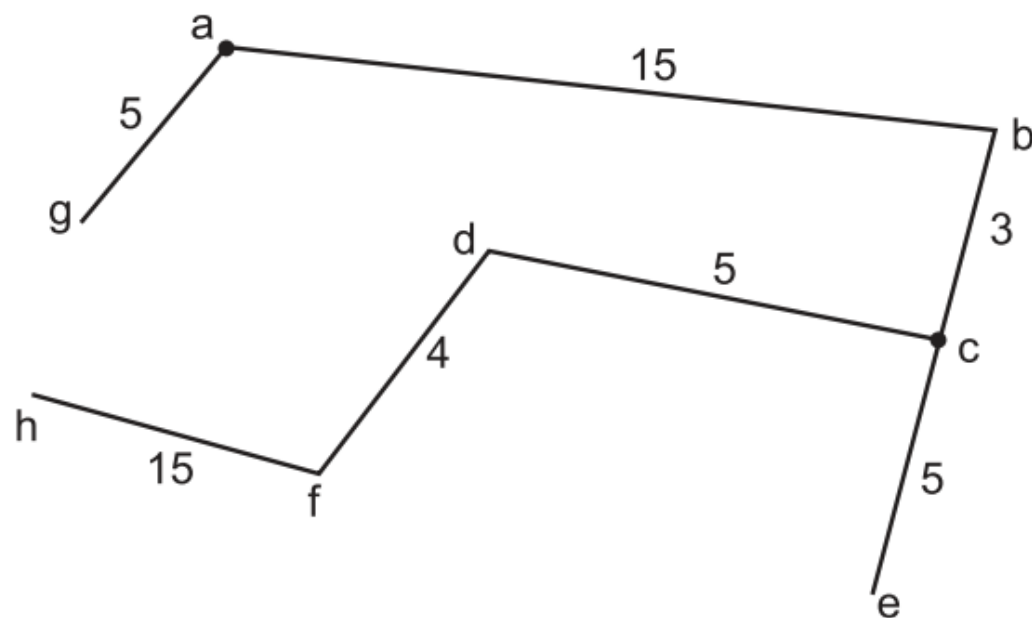
- (ii) Add the next edges $\{d, f\}$, $\{a, g\}$, $\{c, d\}$, $\{c, e\}$.



- (iii) Then we have options : we may choose only one of $\{a, b\}$ and $\{a, d\}$ because selection of both creates a circuit. Suppose that we choose $\{a, b\}$



(iv) Similarly we choose only one of $\{f, h\}$ and $\{g, h\}$. Suppose we choose $\{f, h\}$. then



(v) Edge $\{e, f\}$ and $\{f, g\}$ creates a circuit. Thus, this is a MST.

So, minimal cost for construction of this tree is

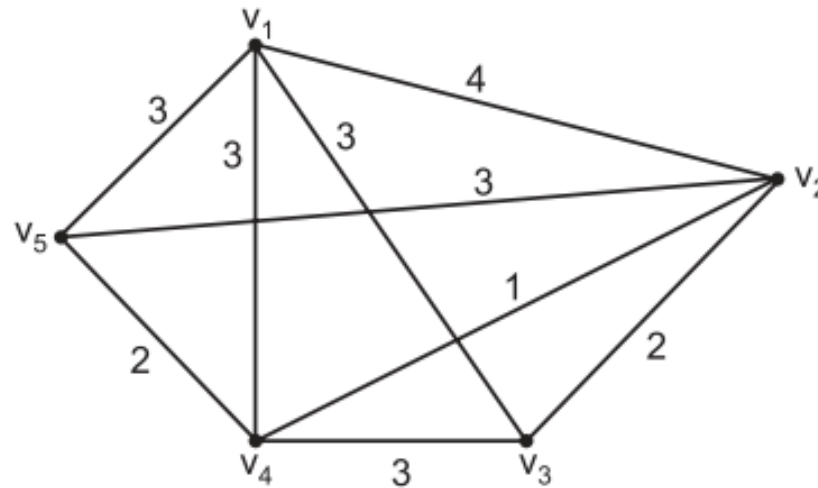
$$5 + 15 + 3 + 5 + 5 + 4 + 15 = 52$$

(B) Prim's Algorithm :

- (i) Draw n isolated vertices and label them $v_1, v_2, \dots, v_n$.
- (ii) Tabulate the given weights of the edges of G in an $n \times n$ table. Set the weights of non-existent edges as very large.
- (iii) Start from vertex v_1 and connect it to its nearest neighbour, *i.e.*, to the vertex which has the smallest entry in row 1 of the table, say v_k .
- (iv) Consider v_1 and v_k as one subgraph and connect this subgraph to its closest neighbour *i.e.*, to a vertex other than v_1 and v_k that has the smallest entry among all entries in row 1 and k . Let the new vertex is v_i .
- (v) Consider the tree with vertices v_1, v_k and v_i as one subgraph and continue the process
- until all n vertices have been connected by $n - 1$ edges.

Example

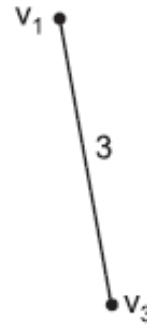
- Find the minimal spanning tree (MST) of the weighted graph shown in figure using Prim's algorithm.



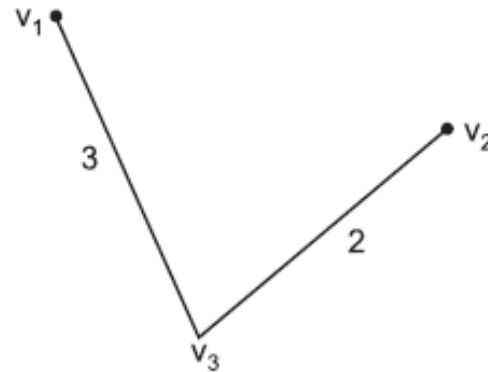
- First, we construct the nxn table from the graph:

	v_1	v_2	v_3	v_4	v_5
v_1	—	4	3	3	3
v_2	4	—	2	1	3
v_3	3	2	—	3	∞
v_4	3	1	3	—	2
v_5	3	3	∞	2	—

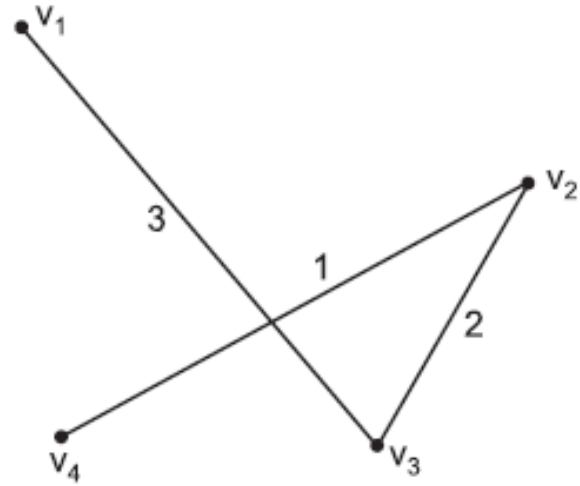
SOLUTION. (i) We choose the vertex v_1 . Edge with smallest weight incident on v_1 is (v_1, v_3) So we choose (v_1, v_3)



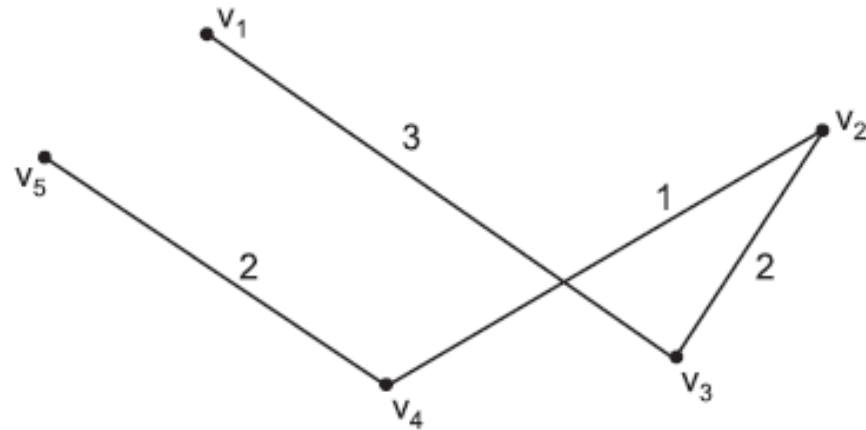
(ii) Now $w(v_1, v_2) = 4$ $w(v_1, v_4) = 3$ $w(v_1, v_5) = 3$ $w(v_3, v_2) = 2$ and $w(v_3, v_4) = 3$. We choose the edge (v_3, v_2) since it is minimum.



(iii) Again $w(v_1, v_5) = 3$, $w(v_2, v_4) = 1$ and $w(v_3, v_4) = 3$. we choose the edge (v_2, v_4) .



(iv) Now we choose the edge (v_4, v_5) .



This is the minimal spanning tree and the minimal cost.

$$= 3 + 1 + 2 + 2 = 8$$